

Scalable Multi-Robot Coordination:

Integrating LCBA Task Allocation with Decentralized Path Planning

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The Problem Context

Context: Modern warehouse fleets require dynamic Multi-Robot Task Allocation (MRTA) to operate efficiently.

The Flaw: Consensus algorithms (CBBA, GCBBA) assume global connectivity or static links.

The Reality: Range-limited radios, obstacles, and agent mobility produce dynamic partitions. Baselines can fail to reach consensus and stall operation.

The Solution: LCBA

The **Local Consensus-Based Bundle Algorithm (LCBA)** relaxes the global connectivity assumption.

- **Component-local consensus:** Bidding and consensus operate within each current connected component.
- **Dynamic resolution:** Time-stamped allocations resolve conflicts on reconnection.
- **Event-driven replanning:** Supports continuous task arrivals without full a priori knowledge.

System Architecture

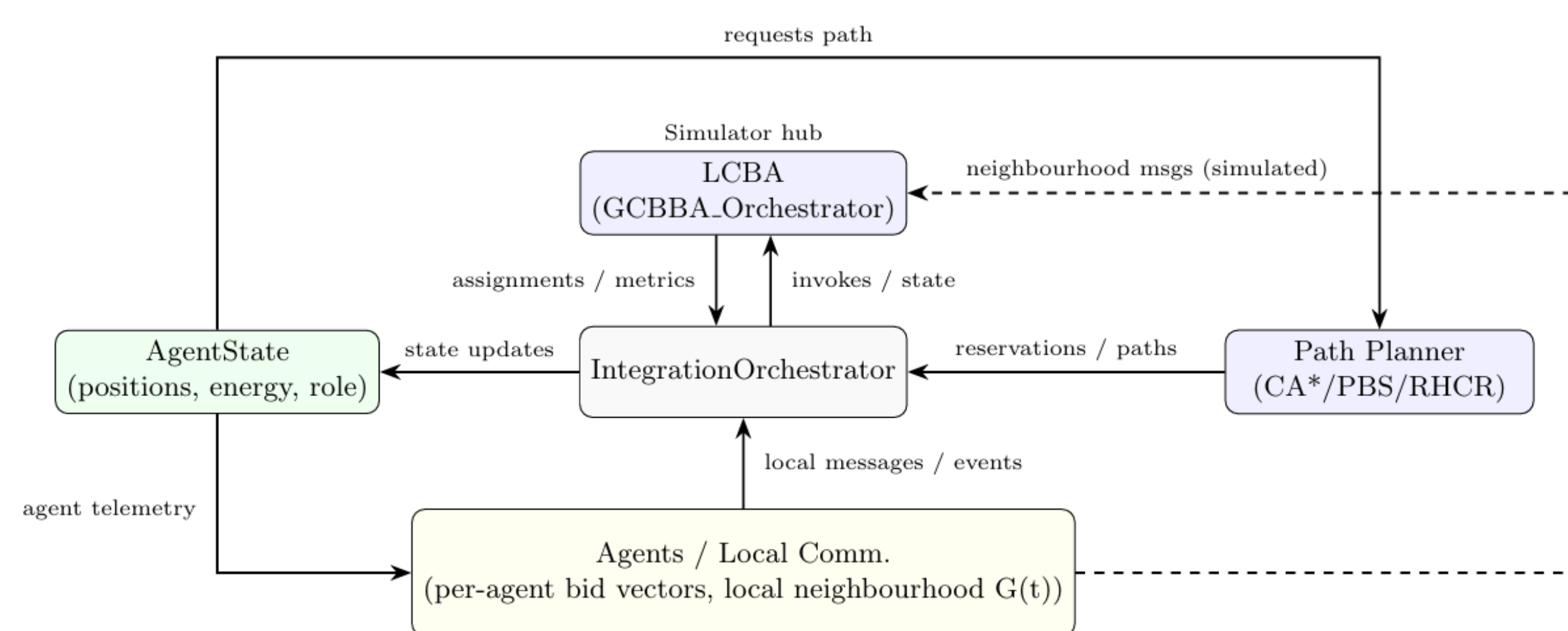


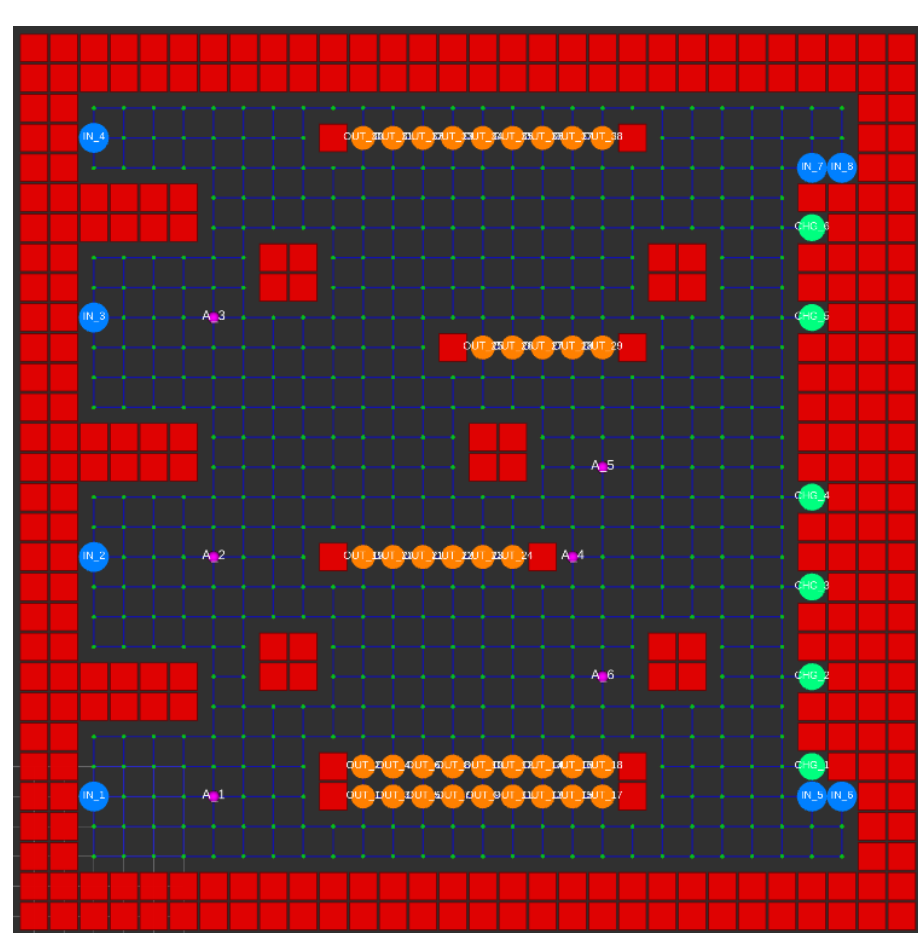
Figure: System Architecture integrating LCBA with role-based planners.

Design: LCBA is logically decentralized, integrating with role-based path planners (RHCR for task execution, CA* for charging) to ensure collision-free execution.

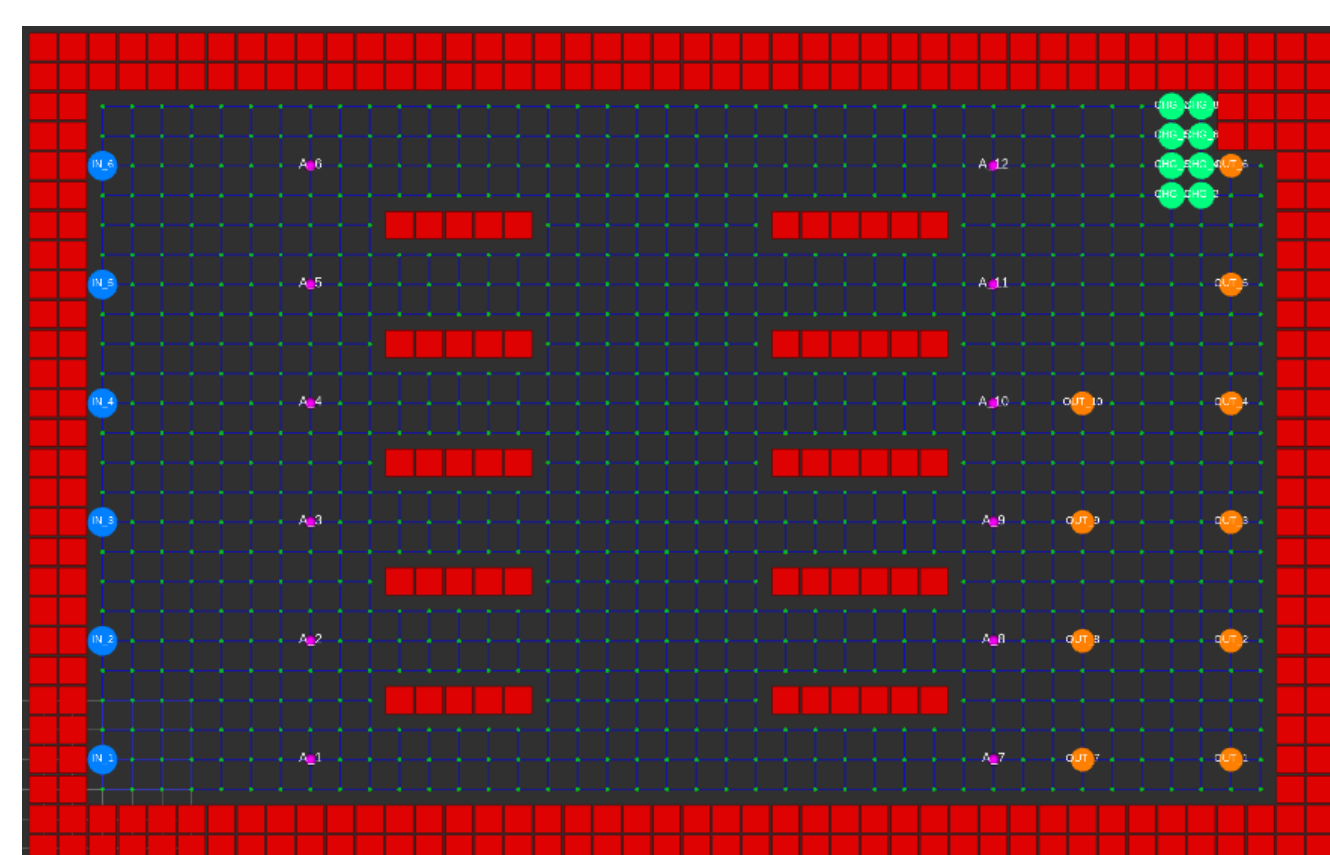
Experimental Setup

Developed a discrete-time gridworld simulator to evaluate algorithmic limits.

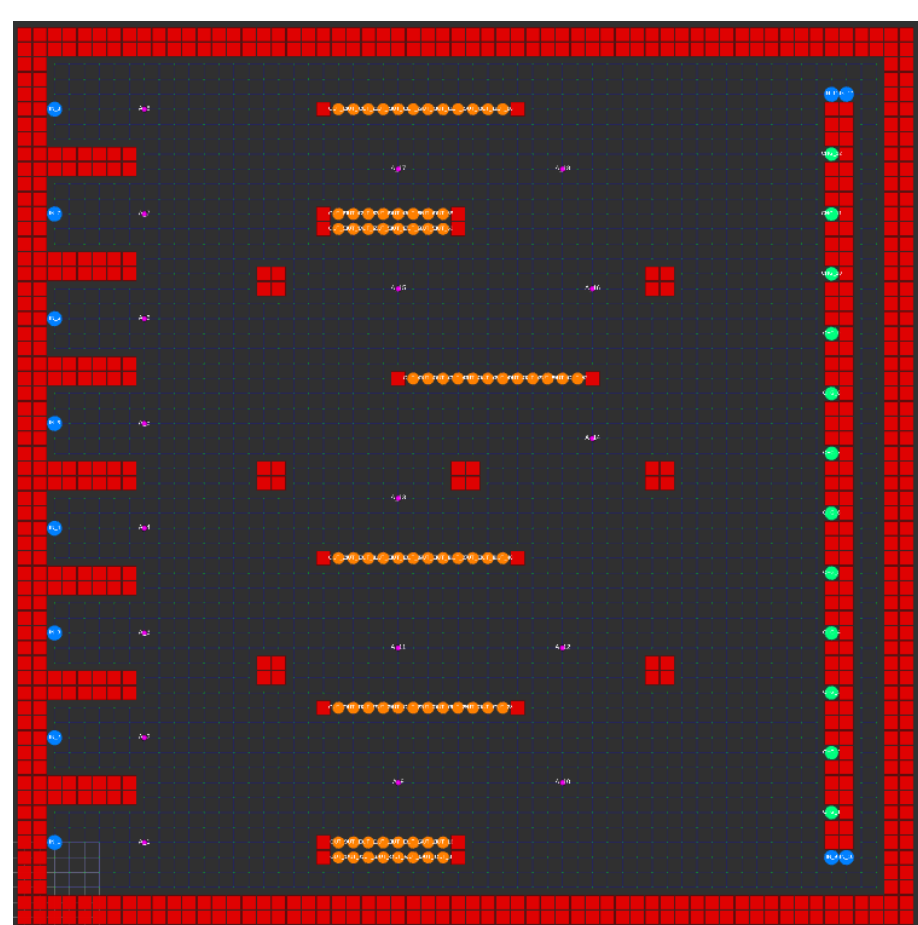
- **Scale:** 432 experimental runs across 6 maps, up to 200 agents.
- **Evaluation:** Batch (makespan) and steady-state (throughput) regimes.



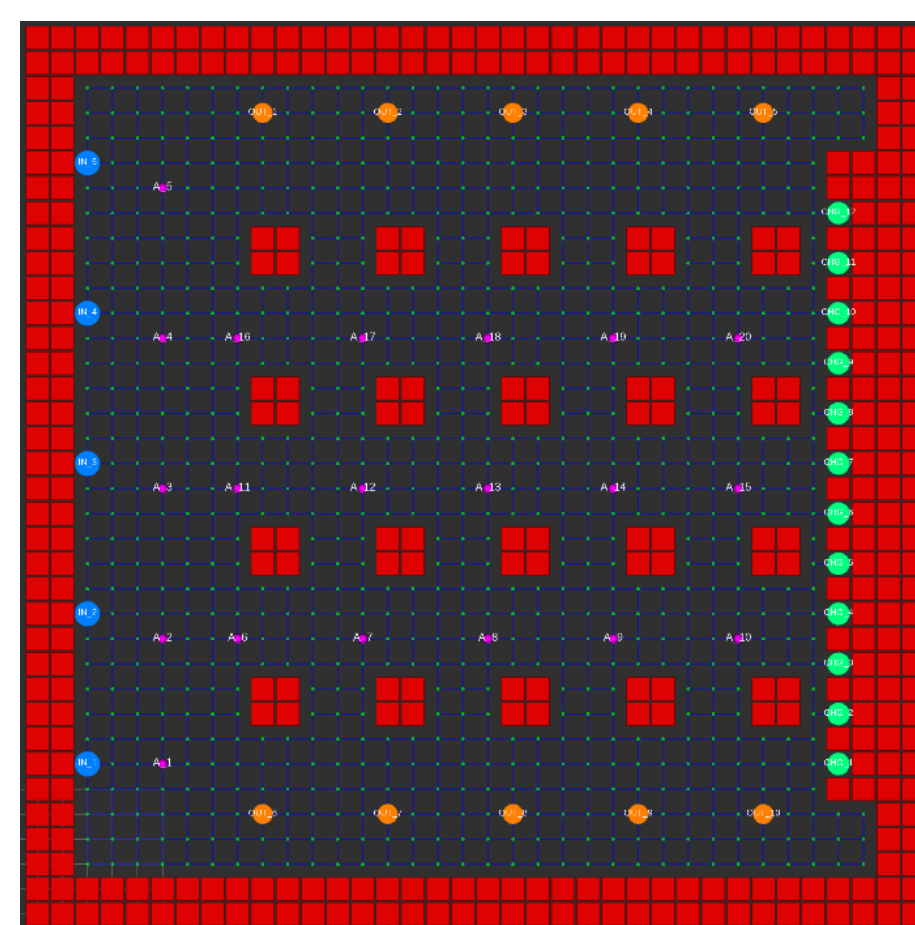
Warehouse Small
30×30, 6 agents



Crossdock
44×28, 12 agents



Warehouse Large
60×60, 18 agents



Kiva
36×36, 20 agents

Conclusion & Future Work

Conclusion: LCBA successfully bridges the gap between mathematically rigorous task allocation and the messy reality of physical robot deployments. By localizing consensus, fleets can operate continuously despite extreme network fragmentation.

Future Work: Integrating fully decentralized collision avoidance and deploying the framework onto physical warehouse robots.

Batch Regime: Communication Robustness

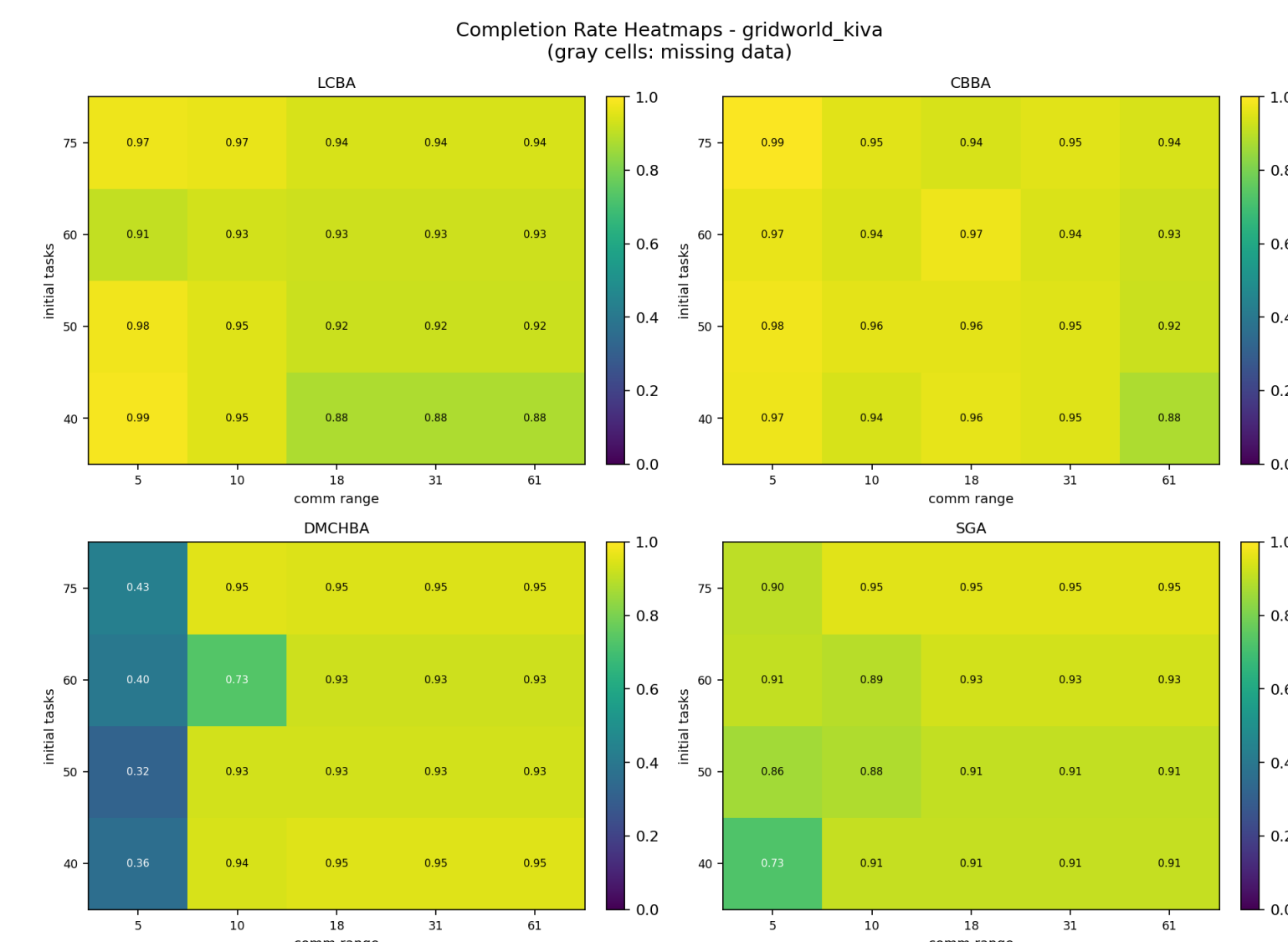


Figure: Batch completion rate heatmap across algorithms (Kiva, 36×36, 20 agents).

- **Robustness:** LCBA maintains high completion across all tested comm ranges and task loads. CBBA is similarly robust, handling network fragmentation gracefully.
- **SOTA Failure:** DMCHBA (SOTA) drops sharply at minimum range. Its per-component Hungarian solves multiply in cost as the network fragments.
- **Clean signal:** No missing data (no wall-clock timeouts): every cell reflects a completed run, making the contrast unambiguous.

Steady-State Regime: Throughput Under Fragmentation

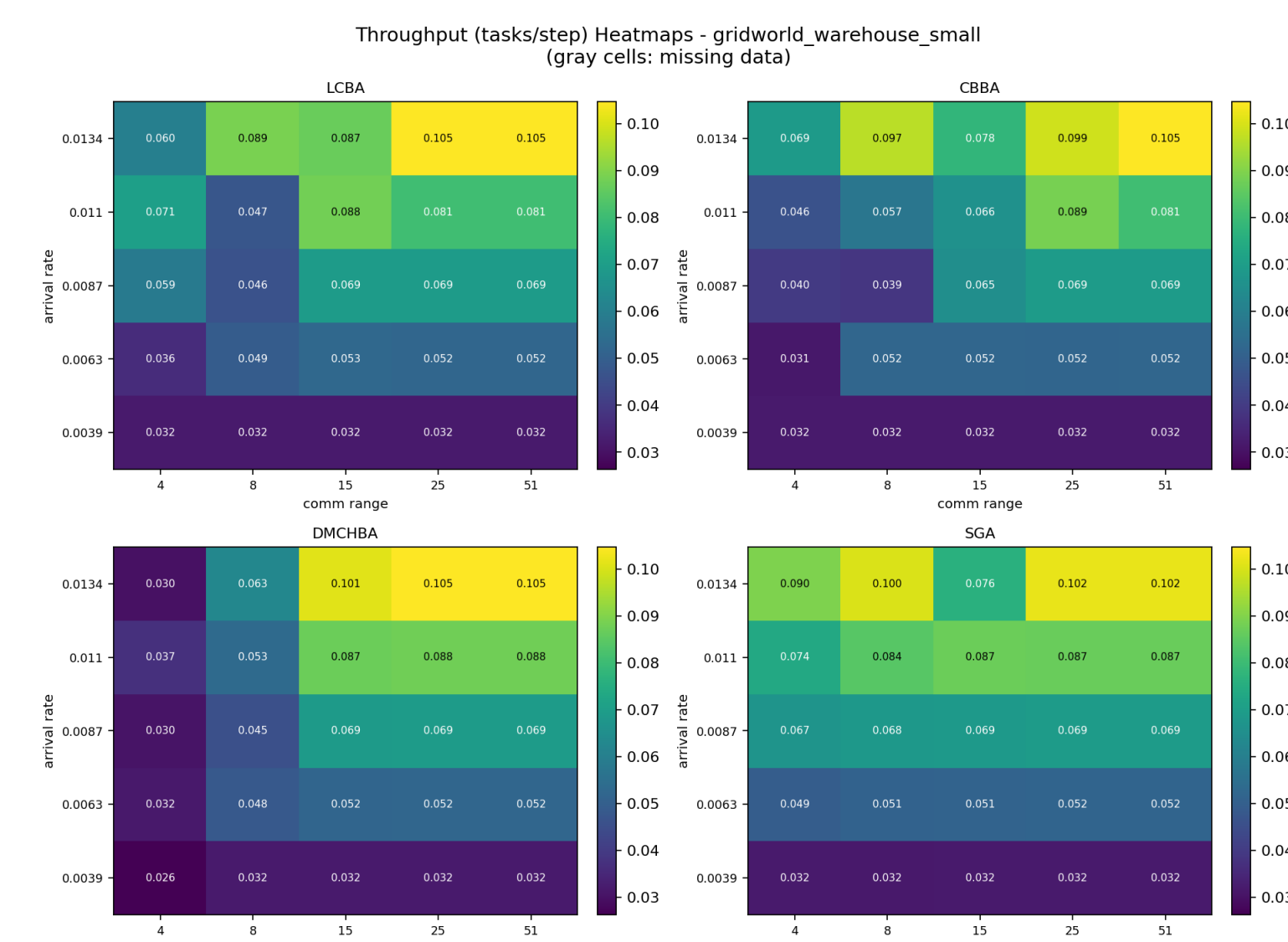


Figure: Steady-state throughput heatmap across algorithms (Warehouse Small, 30×30, 6 agents).

- **Robustness:** LCBA maintains viable throughput across all arrival rates and comm ranges: no stalls, no missing cells, even at minimum range (4 grid units).
- **SOTA outperformed:** At minimum comm range, LCBA outperforms CBBA and DMCHBA (SOTA) in throughput.
- **Convergence:** At full connectivity, all methods reach the same throughput ceiling. LCBA imposes no clear overhead when communication is unrestricted.

Allocation Efficiency — Empirical Evidence

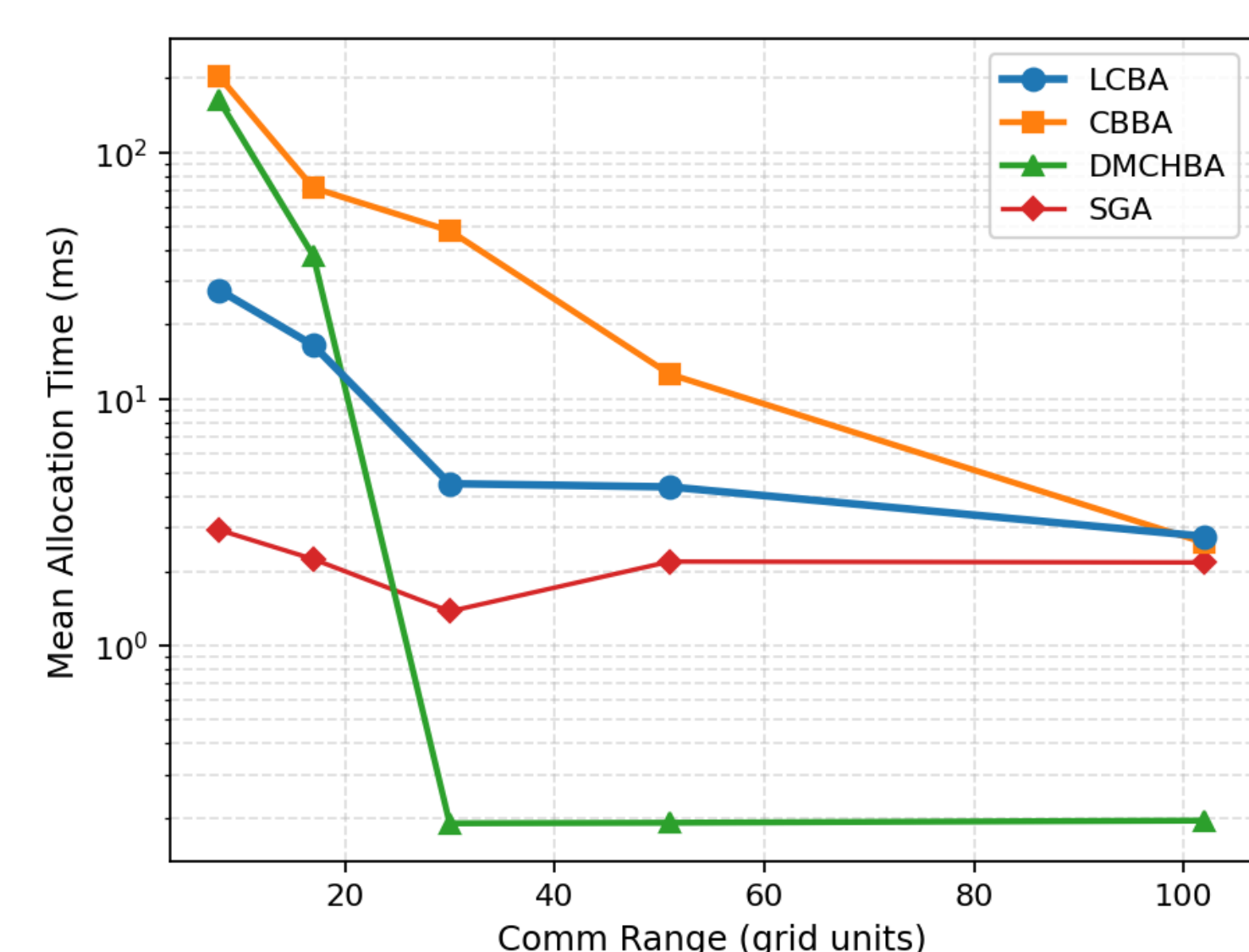


Figure: Mean allocation time vs. communication range (Warehouse Large, 60×60, 18 agents).

- **LCBA vs. CBBA:** At minimum comm range, LCBA is substantially faster than CBBA. LCBA terminates early within each component, while CBBA runs all $N_{\min} \times D$ rounds regardless.
- **LCBA vs. DMCHBA:** LCBA is substantially faster than DMCHBA. DMCHBA clones the full task list and runs Hungarian independently per component, multiplying overhead.
- **Scales with fleet size:** LCBA outperforms CBBA at minimum comm range across all tested fleet sizes. At 18 agents, CBBA runs twice as many consensus rounds (31.6 vs. 15.8), producing 3× higher allocation time (47 ms vs. 15 ms).